

# Deleting an Item from Azure Cosmos DB using .NET SDK

## Summary:-

In this lab, you will use the Azure Cosmos DB .NET SDK to delete a document from an Azure Cosmos DB container. You will identify the document using its **ID** and **Partition Key**, execute the application, and verify that the item has been removed successfully.

## Let's Start:-

After completing this lab, you will be able to:

- Delete an item from an Azure Cosmos DB container using the .NET SDK.
- Identify a document using its **ID** and **Partition Key**.
- Execute the application to remove a document.
- Verify that the document has been deleted successfully.

## Lab Steps:-

### Step 1 – Prepare the Azure Cosmos DB .NET Project

- **Reference:** This lab uses the .NET console application, Azure Cosmos DB account, database, and container created in the previous lab.
- For that **create** Azure Cosmos DB account by selecting **Azure Cosmos DB for NoSQL** from the recommended APIs.
- Configure:  
**Workload type:** Learning, **Region:** EAST US/ EAST US 2, **Capacity Mode:** Provisioned Throughput
- Enable **Apply Free Tier Discount**. Click **Next**.
- Ensure Geo-Redundancy and Multi-region Writes is **Disabled**. Click **Next**.
- Select **Enable access from all networks**. Click on **Review + Create** to create.
- From the left pane, select **Keys**.
- Copy: **URI** and **Primary Key**
- Open **Visual Studio Code**. Create a new project folder.
- Press **Ctrl + Shift + P** to open the **Command Palette**.
- Select **Show and Run Commands > .NET: New Project > Console App**(Rename it as cosmosdb-app) > **Default Directory > .slnx > Create Project**.
- **Open** the integrated terminal. Run:  

```
cd .\cosmosdb-app\  
dotnet add package Microsoft.Azure.Cosmos  
dotnet add package Newtonsoft.Json
```
- Create a new class file named **Order.cs**.
- Open the following project files:

**Program.cs**

**Order.cs**

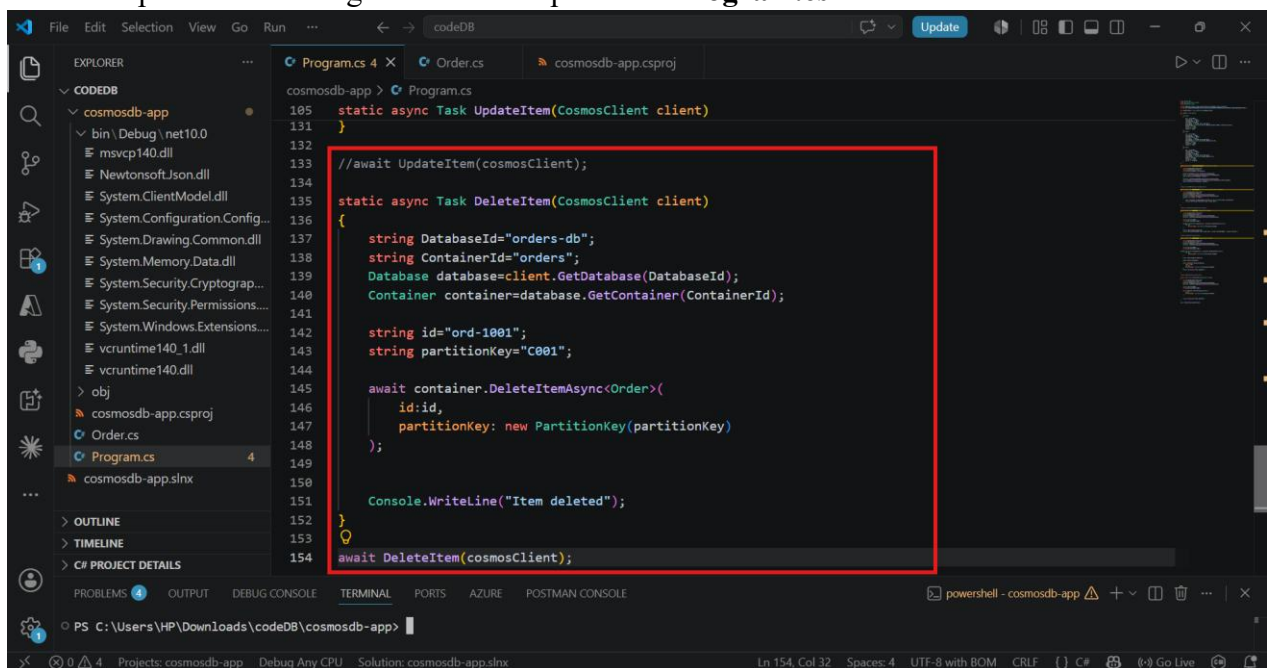
**cosmosdb-app.csproj**

**cosmosdb-app.slnx**

- Replace the existing content with the **provided files**.
- In **Program.cs**, **Replace** Endpoint with **URI** and Key with **Primary Key**.
- **Save** all the files. **Run** in terminal:  
dotnet run  
dotnet build
- This step creates a .NET console application and prepares it to insert items into the Azure Cosmos DB container.

## Step 2 – Update the Application Code

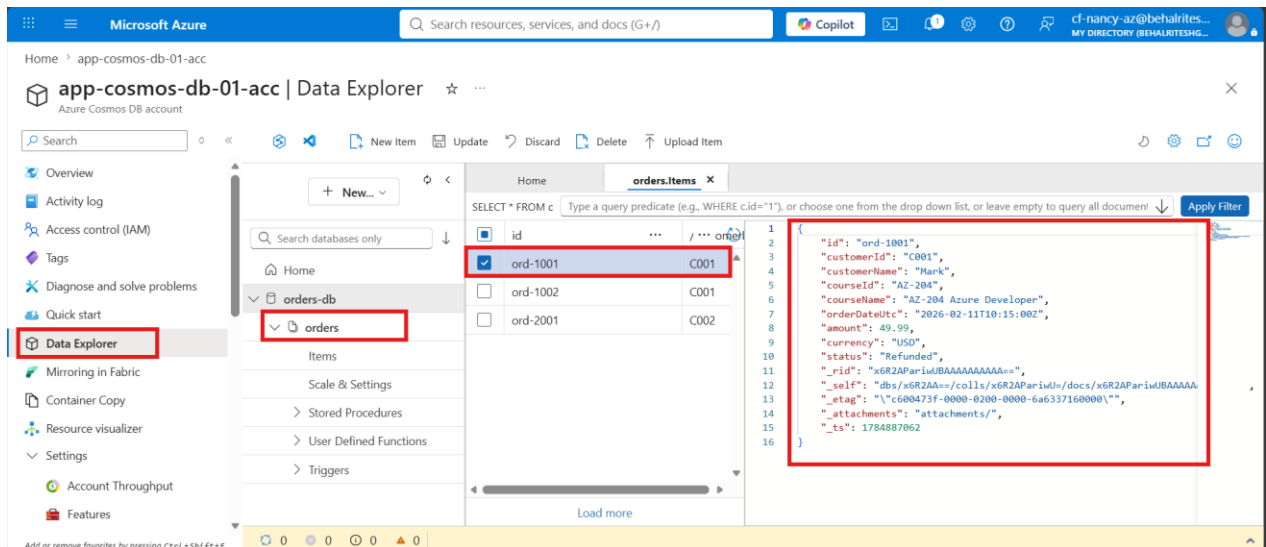
- Open **Program.cs**.
- Replace the existing code with the provided **Program.cs** file.



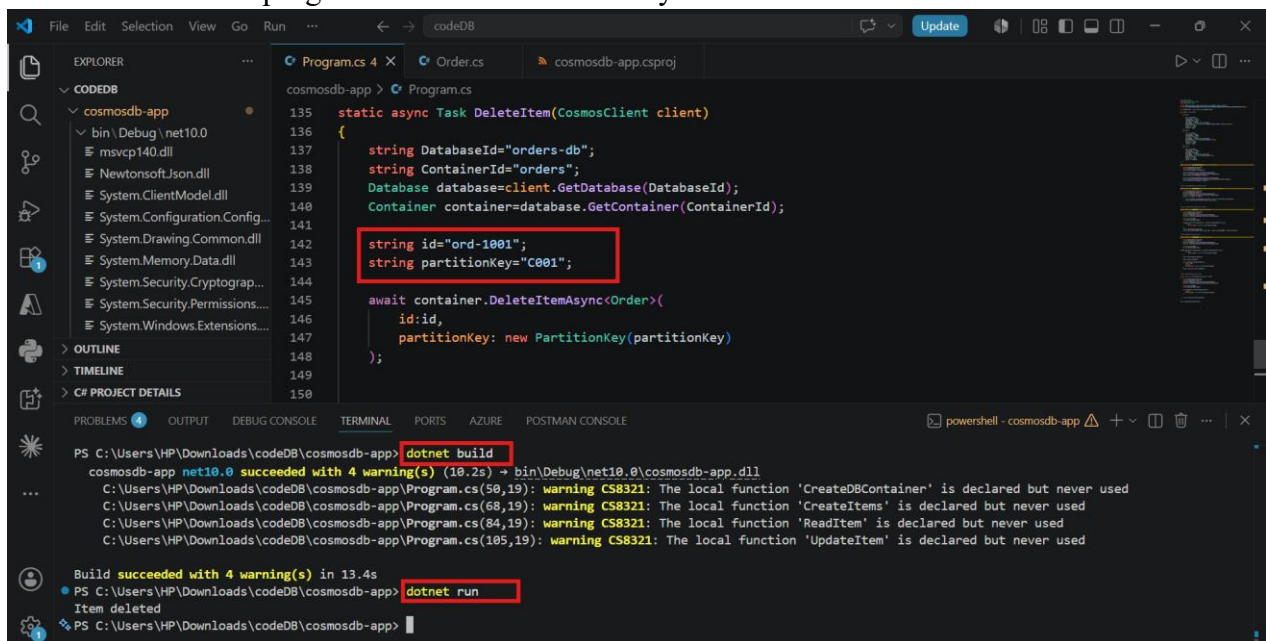
- **Save** the file.
- This step updates the application to delete an item from the Azure Cosmos DB container using the .NET SDK.

## Step 3 – Build and Run the Application

- Verify that the document with **Order ID: ord-1001** is present before deleting it.



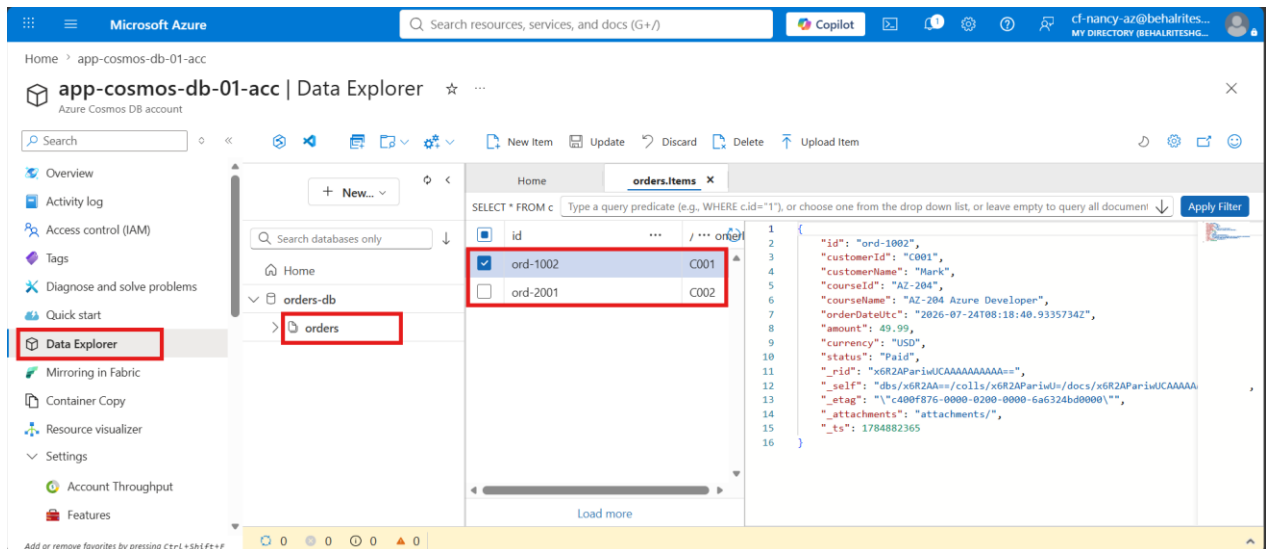
- Open the integrated terminal.
- **Build and run** the application:  
dotnet build  
dotnet run
- Wait for the program to execute successfully.



- This step deletes the specified document from the Azure Cosmos DB container.

## Step 4 – Verify the Deleted Item

- Open the **Azure Portal**. Navigate to **Azure Cosmos DB**.
- Open **Data Explorer**. Expand the **orders-db** database and the **orders** container.
- Select **Items**. Click **Refresh**.



- Observe that the total number of items in the container has decreased.
- This step confirms that the item has been successfully deleted from the Azure Cosmos DB container.

## Verification

Verify the following:

- The application builds and runs successfully.
- The specified document is deleted using its **ID** and **Partition Key**.
- The deleted document no longer appears under **Items**.
- The total number of documents in the container is reduced.
- The delete operation completes without errors.